Final Pitch Script

DOSM Datathon 2026 Final · Team Nexora (UTAR) · How does geopolitical risk affect tourist arrivals to Malaysia? · Deck v5, 12 slides + 4 backups

#	Slide	Speaker	Time	Length
1	Cover	A	0:00 – 0:25	25 s
2	Tourism matters to Malaysia	A	0:25 – 1:05	40 s
3	Test every market, then forecast ahead	A	1:05 – 1:25	20 s
4	Four steps, one rule	B	1:25 – 2:15	50 s
5	The answer is both	B	2:15 – 2:50	35 s
6	Drops and rises cancel out	B	2:50 – 3:10	20 s
7	LightGBM had the smallest error	B	3:10 – 3:30	20 s
8	Most accurate 1–2 months ahead	B	3:30 – 3:50	20 s
9	Live demo: four questions	B	3:50 – 6:40	2 min 50 s
10	Useful, on-theme and realistic	A	6:40 – 7:30	50 s
11	Three things we found	A	7:30 – 8:00	30 s
12	Close	A	8:00 – 8:20	20 s
	Total		**about 8:20**	**1 min 40 s spare**

> Times assume a calm pace of about 140 words a minute. The 1 minute 40 seconds of spare time covers handovers, switching to Power BI, slow moments in the demo and nerves. Speak slowly and clearly. If you are running late, shorten the demo first, never the findings. Orange italic lines are actions, not words to say. ▶ *Click* shows the next item on the same slide; ▶ *Next slide* moves on. Speaker A presents slides 1–3 and 10–12; Speaker B presents slides 4–9.

Slide 1 · Cover

Speaker A · 0:00 – 0:25

Good morning, judges. We are Team Nexora, four students from UTAR, with our advisor Dr. Kelwin Tan.

Our question is simple: how does geopolitical risk, like wars, conflicts and political tension, affect the number of tourists coming to Malaysia? To answer it, we used official data and machine learning on Malaysia's top 20 source markets.

▶ *Next slide*

Slide 2 · Tourism matters to Malaysia

Speaker A · 0:25 – 1:05

Tourism is very important to Malaysia. It makes up 15.9 percent of our GDP and supports 3.7 million jobs. So when risk goes up somewhere in the world, it can reach our hotels, airlines and homestays.

▶ *Click* But here is the problem. If you look at Malaysia as one national number, the link between risk and tourist arrivals is only plus 0.08. That is almost zero.

▶ *Click* Past research, by Lee and others in 2021, also found no link at the national level. And nobody had checked each country separately.

▶ *Next slide*

Slide 3 · Test every market, then forecast ahead

Speaker A · 1:05 – 1:25

So our objective was to test how geopolitical risk links to arrivals from each of our top 20 markets, and to forecast arrivals one to four months ahead.

Just think about this for a moment: when risk goes up, do tourists come less, come more, or both? My teammate will now explain how we found out.

Optional: click A, B and C on the slide as you say less, more, both; each badge pulses.

▶ *Next slide · hand over to Speaker B*

Slide 4 · Four steps, one rule

Speaker B · 1:25 – 2:15

Thank you. We did this in four steps.

▶ *Click* Step one, we collected official open data: about 69 thousand daily rows, which we turned into 2,300 country-months.

▶ *Click* Step two, we dated every number by the day it is actually published, not the day it happened.

▶ *Click* Step three, we used Ridge regression to measure the link with each market, and a LightGBM model to forecast one to four months ahead.

▶ *Click* Step four, we tested it fairly, with five rolling tests on past years and a COVID stress test, and put the results in a Power BI dashboard.

▶ *Click* Our main rule: the model only learns from numbers that were already published, so it never sees the future.

▶ *Next slide*

Slide 5 · The answer is both

Speaker B · 2:15 – 2:50

Now our first finding. Back to the question: does risk push tourists away, or bring more in? Our answer is: both.

When we look at each country one by one, we see three groups.

▶ *Click* For 16 of our 20 markets, like Nepal, South Korea and Germany, higher risk goes with fewer tourists.

▶ *Click* For 3 markets, Singapore, Indonesia and China, higher risk goes with more tourists.

▶ *Click* And for Thailand, there is almost no change.

▶ *Next slide*

Slide 6 · Drops and rises cancel out

Speaker B · 2:50 – 3:10

Here is our key insight. The drops and the rises cancel each other out. That helps explain why the national link is only plus 0.08.

▶ *Click* One national number hides what is really happening, so risk must be watched market by market.

We are careful here: these are patterns, not proof that risk causes the change.

▶ *Next slide*

Slide 7 · LightGBM had the smallest error

Speaker B · 3:10 – 3:30

Our second finding is about forecasting. We tested every model on past data, as if it were live.

Our LightGBM model had the smallest error, about 16 thousand arrivals per market per month. Overall, it beat Ridge, repeating last month, and repeating last year.

▶ *Next slide*

Slide 8 · Most accurate 1–2 months ahead

Speaker B · 3:30 – 3:50

Forecasts are most reliable one and two months ahead, with error under 12 thousand. At three and four months, the error about doubles.

▶ *Click* And one honest result: during COVID, simply repeating last month beat our model, because a shock like that is not in the past data our model learns from.

▶ *Next slide*

Slide 9 · Live demo: four questions

Speaker B · 3:50 – 6:40

Now let me show you our dashboard. It has four pages, listed on the left.

Alt+Tab to Power BI. It should already be open on Malaysia Tourism Overview, full screen, both slicers set to All.

Overview (30 s). *Point at the four cards.* This page is the big picture for our 20 markets. The latest month, May 2026, had 1.96 million arrivals, 3.1 percent below May last year. But the year so far is up 0.9 percent, at about 10 million. *Point at Top 5 Source Countries.* This year, Singapore, China and Indonesia are our biggest markets. *Point at Annual Arrivals and YoY Growth.* And here you can see the COVID gap in 2020 and 2021, and the sharp jump in growth when borders reopened in 2022.

1. Which markets matter? (30 s). *Click Market Analysis.* Across our whole data period, ASEAN countries bring almost three in four of our arrivals, led by Singapore, Indonesia and China. *Click the Non-ASEAN part of the donut.* When I click Non-ASEAN, the chart shows only those markets, led by China. But being big is not the same as being sensitive to risk. *Click blank space to clear.*

2. What moves arrivals? (40 s). *Click Tourism Drivers.* Here we put arrivals next to oil prices, the exchange rate and global geopolitical risk, plus Malaysia's own risk index. *Point at the last blue bar.* One note: the last bar is 2026 so far, January to May only, so it is not a drop. Over the same five months, arrivals are up 0.9 percent. *Point at Tourist Arrivals vs Global Geopolitical Risk.* Look at this one. Since 2022, global risk and total arrivals have both gone up, because tourism was recovering. When all markets are added together, that recovery hides the link, which is why we test each market on its own. *Set Source Country to Singapore, then back to All.* You can do the same for any market. Each chart carries a short note, because these are associations, not causes.

3. What do we forecast? (35 s). *Click Forecast and point at Forecast Arrivals 2026.* For our 20 markets, the model forecasts about 2.3 million arrivals a month in June and July 2026, easing to about 2.1 million by September. *Set source_country_name to Singapore, then back to All.* You can also see the forecast for one market. These are single best estimates; adding ranges is our next step. *Point at Feature Importance.* On the right are the inputs the model uses most: recent arrivals and last year's season come first, and global risk is sixth.

4. How accurate is it? (30 s). *Point at Model Accuracy | Avg MAE (K). Do not point at the 16.8% card; if asked, use the Q&A answer.* Tested on past years as if it were live, LightGBM had the lowest average error, about 16 thousand arrivals per market per month, ahead of Ridge and the two simple methods. *Point at COVID Stress Test.* But during COVID, simply repeating last month had the lowest error, and we show that openly.

Alt+Tab back to the slides and hand over to Speaker A. If Power BI freezes, click Demo backup and talk through the screenshots instead.

▶ *Next slide · hand over to Speaker A*

Slide 10 · Useful, on-theme and realistic

Speaker A · 6:40 – 7:30

So why does this matter?

▶ *Click* First, who could use it. Tourism Malaysia could decide which markets to promote in, and when. DOSM could add a market-level view next to the national number. And hotels, airlines and homestays could plan their staff earlier.

▶ *Click* Second, it fits this year's theme, machine learning for sustainable tourism. It links to SDG 8.9 on tourism jobs and local culture, SDG 12.b on tools to monitor tourism, and the National Tourism Policy 2020 to 2030.

▶ *Click* Third, it is realistic. It uses only open, official data, it updates every month with one command, and it can grow to more markets and states.

To be clear, we show how it could support decisions. We have not measured that impact yet.

▶ *Next slide*

Slide 11 · Three things we found

Speaker A · 7:30 – 8:00

To conclude, we found three things. Each market reacts differently when risk rises.

▶ *Click* Looking market by market shows what the national number hides.

▶ *Click* And our model forecasts best one to two months ahead.

▶ *Click* We are also honest about our limits. These are patterns, not causes. Sudden shocks like COVID cannot be predicted. And we only cover 20 markets, so forecast ranges are our next step.

▶ *Next slide*

Slide 12 · Close

Speaker A · 8:00 – 8:20

So, how does geopolitical risk affect tourism in Malaysia? For 16 of our 20 markets, higher risk goes with fewer tourists, but not for all of them. That is why looking market by market matters.

▶ *Click* Terima kasih. We are happy to take your questions.

Q&A: click Accuracy, Markets, Data & design or Demo on this slide to open backups 13–16; each backup has a Q&A index button back to slide 12.

Q&A answers (5 minutes)

Short answers in simple English. Backup slides: A for accuracy, B for markets, C for data, D for demo screenshots.

Why do Singapore, Indonesia and China go up when risk goes up?

We don't claim risk causes it. Risk went up in the same years that tourism recovered after COVID, so we think it is mostly the recovery trend.

Is LightGBM really better than Ridge?

It had the lowest overall error, 15,969, and the lowest RMSE. But Ridge won 7 of the 20 test cells, so our choice is reasonable, not a big win.

How do you stop the model from seeing the future?

Every input is delayed by its real publishing time. The model only learns from arrivals that were already published on the forecast date.

What if the arrivals data is published later than you assumed?

We tested a 3-month and a 4-month delay. The best model stayed the same.

How important is geopolitical risk in the model?

It is the 6th most used input. Past arrivals and season matter more, which makes sense.

How would you measure SDG impact?

Our report lists indicators, such as tourism jobs, wages and local SME participation for SDG 8.9, and energy, water, waste and traffic for SDG 12.b.

Why is the 2026 bar so low on Tourism Drivers?

2026 only has January to May so far. Compared with the same five months of 2025, arrivals are up 0.9 percent.

What is the 16.8% on the Forecast page?

It is a weighted percentage error, WAPE: total error divided by total arrivals, across all 20 markets. On our slides we use average error in arrivals, because small markets make percentages jump.

Can it run without your laptop?

Yes. The Power BI file has its data built in, the data file is included, and the Python pipeline updates with one command. Code is on GitHub: github.com/william20808/nexora_dosm.

Backup D (slide 16) holds the demo screenshots if Power BI fails.